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Research Paper

High-dimensional multi-objective optimization of coupled cross-laminated timber walls building using deep learning

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ABSTRACT

This paper introduces a deep learning-based multi-objective optimization framework, applying for advancing the design of the Cross-Laminated Timber Coupled Wall system. While traditional optimization methods often struggle with the curse of dimensionality in high-dimensional problems, the approach proposed in this study employs an autoencoder to effectively reduce the dimensionality of the design space. Subsequently, a neural network establishes a mapping between input variables and latent spaces, with another neural network forming the crucial link between these latent variables and the output responses. The proposed framework is integrated into the design process of a 20-story Cross-Laminated Timber Coupled Wall system, where uncertainties inherent in connection elements are systematically addressed to optimize the structural parameters. The non-dominated sorting genetic algorithm-II is utilized to estimate optimal design variables by minimizing three conflicting objective functions, thereby generating a Pareto front. This optimized design is then bench-marked against three deterministic models with varying coupling beam shear strengths. A two-dimensional numerical model developed in OpenSees facilitates nonlinear time history analysis using 50 bi-directional ground motion records, representative of the seismicity of Vancouver, Canada. The results of this study not only highlight the efficacy of the deep learning-based framework in enhancing the structural integrity and resilience of high-rise timber structures in seismic regions but also significantly contribute to the evolution of computational approaches in structural engineering.

1. Introduction

Real-world complex systems often have several objectives, many of which are mutually exclusive and/or conflicting objectives. Therefore, there is an ongoing in the engineering and research domains about the solution to such multi-objective optimization (MOO) problems. In the presence of uncertainty, MOO provides a set of optimal solutions, called Pareto optimal solutions, by solving the multi-objective functions optimized simultaneously. With respect to all objective functions, none of those Pareto optimal solutions can be said to be superior to any other. The MOO can be formulated as:

$$\begin{aligned}&\text{Minimize/Maximize} \quad \mathbf{Y} = f_i(\mathbf{X}) \qquad i = 1,2,\dots,N_{obj}\\ &\text{Subject to} \quad : \begin{cases} g_k(\mathbf{X}) = 0, & k = 1,2,\dots,K \\ h_l(\mathbf{X}) \le 0, & l = 1,2,\dots,L \\ \mathbf{X}^{(L)} \le \mathbf{X} \le \mathbf{X}^{(U)} \end{cases}\end{aligned} \tag{1}$$

where $\mathbf{Y}$ represents the objective function, which is to be minimized or maximized. Also, N_{obj} denotes the total number of objective functions considered in the multi-objective formulation. The optimization may be carried out in the presence of constraints or without constraints. In Eq. (1), the equality and inequality constraints are denoted by $g_k(\mathbf{X})$ and $h_l(\mathbf{X})$, respectively. Also, the decision variables, $\mathbf{X}$, are bounded between the lower bound, $\mathbf{X}^{(L)}$, and the upper bound, $\mathbf{X}^{(U)}$, respectively.

Over the last few decades, various methods have been proposed to solve such MOO problems. The global criterion method (Miettinen, 1999) converts the multiple objective functions into a single objective function by minimizing the distance between ideal solutions and viable destination areas. Cohon (2004) used a weighted sum method in which a single objective function is formed by taking the weighted sum of all objective functions. Although these methods are simple to use, they cannot produce Pareto optimal solutions when the function is non-convex. Therefore, in recent years, evolutionary algorithms have gained interest. Among them, Multi-objective Genetic Algorithm (MOGA) (Fonseca et al., 1993), Multi-objective Particle Swarm Optimization (MOPSO) (Kennedy and Eberhart, 1995), Pareto Archived Evolution Strategy (PAES) (Knowles and Corne, 2000), Strength Pareto

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Evolutionary Algorithm Version 2 (SPEA2) (Zitzler et al., 2001), Non-dominated Sorting Genetic Algorithm-II (NSGA-II) (Deb et al., 2000), etc. are popular. The efficiency of the different evolutionary algorithms has been investigated by Das and Tesfamariam (2022). Though the above evolutionary algorithms for solving MOO problems have specific advantages, they usually require multiple computational model evaluations to achieve satisfying Pareto optimal solutions with adequate accuracy. When the computational model is complex in nature or has high nonlinearity in the system, the multiple evaluations of the computational model increase the computational budget. In this context, the evolutionary algorithm coupled with the surrogate model is useful (Pech et al., 2019), where the surrogate model approximates the original response surface and the evolutionary algorithm is applied to the response surface obtained from the surrogate model to obtain Pareto optimal solutions. Response surface method (Kaymaz and McMahon, 2005), radial basis function (RBF) (Chen et al., 2022), Kriging (Tondut et al., 2022), polynomial chaos expansion (PCE) (Wang et al., 2021), support vector machine (Wu et al., 2023), artificial neural networks (Vamplew et al., 2021) are readily used surrogate models. Regis (2016) used RBF-based NSGA-II for multi-objective and constrained black-box optimization. Zhu et al. (2014) used Kriging-based NSGA-II for solving MOO problems. Kriging-based constrained MOO algorithm was proposed by Singh et al. (2014). Other uses of surrogate models for optimization problems include Das et al. (2020, 2021b,a).

The accuracy of the MOO algorithm is dependent on the accuracy of the surrogate models. These surrogate models are effective in low-dimensional space, while it will be difficult to construct a surrogate model in high-dimensional space as it suffers from the curse of dimensionality. Therefore, the dimension reduction technique is one of the alternatives in which the high-dimensional space transforms into a low-dimensional space, and subsequently, a surrogate model is built on the low-dimensional space. Most dimension reduction techniques are categorized into two classes, i.e., feature selection (García-Torres et al., 2023) and feature extraction (Hira and Gillies, 2015). The feature selection method determines the input variables that have the most impact (Saltelli et al., 2008). On the other hand, the feature extraction strategy converts the high-dimensional space to a reduced space. The primary drawback of the feature selection approach is that it is ineffective when every input variable is about equally important to the amount of interest. The feature extraction approach overcomes this limitation as it is a combination of all input variables. There are different methods used for dimension reduction of a problem, such as Principal Component Analysis (PCA) (Vidal et al., 2005), Linear Discriminant Analysis (LDA) (Duda et al., 2000), Karhunen–Loeve expansion (Ghanem and Spanos, 2003; Diez et al., 2015), t-Distributed Stochastic Neighbor Embedding (t-SNE) (Van der Maaten and Hinton, 2008), active subspace (Constantine, 2015), and high-dimensional model representation (HDMR) methods (Chowdhury and Rao, 2009), etc. Deep learning methods (Sabique et al., 2023) have gained much attention because they can easily transform high-dimensional data into low-dimensional space. Accordingly, in this study, a novel deep learning-based framework is proposed where the autoencoder (Zhang et al., 2024) is used to reduce the dimension of a problem. Then a mapping is established between the input and latent variables using a neural network. Another neural network is then used to form a relationship between the latent variables and the output responses. The non-dominated sorting genetic algorithm-II is used to estimate optimal design variables by minimizing three conflicting objective functions, which results in the formation of the Pareto front.

The utility of the proposed framework is demonstrated in the design of the Cross-Laminated Timber Coupled Wall (CLT-CW) system. The system features CLT balloon shear walls, steel coupling beams with replaceable shear links, and buckling-restrained brace (BRB) hold-downs (Tesfamariam et al., 2021b). As part of the deterministic design procedure, the authors of this manuscript contributed to its design methodology (Tesfamariam et al., 2021b; Teweldebrhan and Tesfamariam, 2022), assessed the seismic ductility-modification factors (Teweldebrhan et al., 2022), and evaluated its seismic resilience through FEMA P-58 methodologies (You et al., 2023). The significant interest and potential of the CLT-CW system are further evidenced by recent experimental studies, such as the cyclic experimental tests conducted by Moerman et al. (2022, 2024). A range of analytical and numerical explorations also exist, such as those by Pei et al. (2017), who integrated CLT floors as coupling elements, and Tesfamariam et al. (2021a) and Dowden and Tatar (2019), who investigated the use of reinforced concrete and replaceable structural fuses in coupling beams, respectively. Moreover, BRB hold-downs, another key feature of the system, have been extensively discussed numerically (e.g., Tesfamariam et al. 2021a, Busch et al. 2022), tested experimentally (e.g., Araújo et al. 2023), and implemented in practice (e.g., Zimmerman et al. 2020). Existing studies on the CLT-CW system (e.g., Tesfamariam et al. 2021b), however, considered deterministic models. Building on the outcomes and limitations of the existing studies, this paper extends the research to incorporate uncertainty in modeling parameters and utilizes MOO to refine the design solutions of the system.

The outline of the study is structured as follows. Section 2 describes the proposed deep learning-based multi-objective optimization method. Section 3 presents the layout of the CLT-CW building model and the design details, such as connection details and seismic design of the CLT-CW system, along with a brief description of seismic hazards and GM selection. The problem formulation with the objective functions and the numerical results using the proposed method are presented in Sections 4 and 5, respectively. Finally, Section 6 presents the concluding remarks.

2. Deep learning based multi-objective optimization

In this section, a brief description of the autoencoder for dimension reduction is provided. The details of the proposed deep learning-based multi-objective optimization are described.

2.1. Autoencoder based dimension reduction

An autoencoder is a type of artificial neural network used in unsupervised machine learning and used for dimensionality reduction tasks. It is primarily designed for data compression and feature learning. Typically, the autoencoder architecture consists of two main components, which are an encoder and a decoder, as shown in Fig. 1. The first component, the encoder, transforms the input data into a compressed or encoded representation. It typically consists of one or more layers of neurons that reduce the dimensionality of the input data. The middle part of the autoencoder is called the latent space, which contains the compressed representation of the input data. This layer has a lower dimensionality than the input data, forcing the autoencoder to learn a compressed representation. The last part of the autoencoder is the decoder, which is responsible for reconstructing the original input data from the encoded representation. Like the encoder, it also consists of one or more layers of neurons, but it typically expands the dimensionality back to the original input dimension.

In Fig. 1, n inputs are considered, which is $\mathbf{X} = \left[\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n\right]^T$. Also, it is assumed that there are m samples for each variable $\mathbf{X}_i$, $i = 1, \ldots, n$. Therefore, the training data for the autoencoder is expressed as follows

$$\mathcal{D} = \begin{bmatrix} X_1^{(1)} & X_2^{(1)} & \cdots & X_n^{(1)} \\ \vdots & \vdots & \cdots & \vdots \\ X_1^{(m)} & X_2^{(m)} & \cdots & X_n^{(m)} \end{bmatrix} \tag{2}$$

Let the latent space of the autoencoder in Fig. 1 be represented by z. Therefore, a mapping from $\mathbf{X}$ to latent vector z is formed, which is expressed as $z = E_\phi(\mathbf{X})$. It is noted that both the encoder and decoder parts are two different multilayered perceptrons (MLP). $E_\phi(\mathbf{X})$

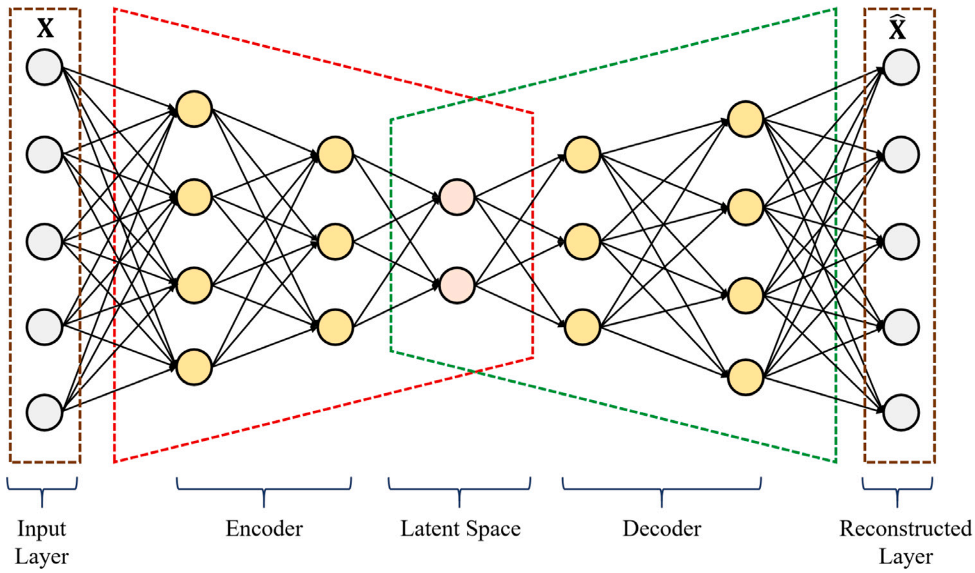


Fig. 1. Schematic diagram of the structure of an autoencoder neural network.

is the output of the MLP for the encoder part, which is expressed as $E_\phi(\mathbf{X}) = \sigma(\mathbf{W}\mathbf{X} + b)$. The activation function of the neural network is expressed as $\sigma(\cdot)$. Also, ϕ is the parameter of MLP for the encoder part, i.e., $\phi = [\mathbf{W}, b]$, where $\mathbf{W}$ and b are the weight and bias of the neural network. Similarly, for the decoder part, which reconstructs the data, $\hat{\mathbf{X}}$ from $\mathbf{X}$, another MLP is formed, which is expressed as $\hat{\mathbf{X}} = F_\theta(z) = F_\theta[E_\phi(\mathbf{X})]$. In Fig. 1, the MLP for the decoder part is expressed as F_θ, which is parameterized by θ. Therefore, $F_\theta(z) = \sigma(\mathbf{W}'z + b')$, where $\theta = [\mathbf{W}', b']$. The training process of an autoencoder involves minimizing the reconstruction error, which is the difference between the input data and the output of the decoder. The reconstruction error can be expressed in the following form

$$\mathcal{L}(\theta, \phi) = \mathbb{E}[d(\mathcal{D}, \mathcal{D}')] \tag{3}$$

where $d(\cdot)$ is the L_2-norm between any sample points. Also, $\mathbb{E}(\cdot)$ denotes the expectation operator. Using the gradient descent technique, the reconstruction error, $\mathcal{L}$, is minimized to find the neural network parameters, θ and ϕ.

2.2. Deep learning-based multi-objective optimization

In this study, a three-step framework is proposed for multi-objective optimization. The first step of the framework is devoted to reduce the high-dimensional data to a low-dimensional space. To do that, an autoencoder is used, whose details are provided in Section 2.1. As our aim is to reduce the dimension, we only use the encoder part of the autoencoder, which establishes a mapping between the input data and latent variables. To train the autoencoder, the input $\mathbf{X}$ and the corresponding outputs $\mathbf{Y}$ are fused as $\mathcal{D} = [\mathbf{X}, \mathbf{Y}]^T$. Using the autoencoder, the training data, $\mathcal{D}$, is changed into the latent space, $\mathbf{z}$, and as a result, the output of the encoder, $\mathbf{z} = [z_1, z_2, \ldots, z_m]$, is obtained. The next step is to establish a mapping between the input and latent variables. For this, a feed-forward neural network (MLP) is chosen that is trained using the input variables $\mathbf{X}$ and the latent variables ($\mathbf{z}$) obtained from the first step, which is expressed as $\mathbf{z} = \mathcal{N}\mathcal{N}_1(\mathbf{X})$. Once the training is finished, a large set of samples of input variables ($\mathbf{X}^p$) are generated using Monte Carlo simulation, which are treated as predicted samples. These predicted samples, $\mathbf{X}^p$, are passed through a trained neural network, i.e., $\mathcal{N}\mathcal{N}_1(\cdot)$, resulting in the obtained predicted latent variables, $\mathbf{z}^p$. The last step is to form a relationship between latent variables and output. To do that, another feed-forward neural network is considered. This network is trained by the latent variables obtained in the first step, $\mathbf{z}$, and the output corresponding to the training samples, $\mathbf{X}$, which can be expressed as $\hat{\mathbf{Y}} = \mathcal{N}\mathcal{N}_2(\mathbf{z})$. Once the training of the neural network is completed, the predicted latent variables obtained in the second step, $\mathbf{z}^p$, are passed through the neural network $\mathcal{N}\mathcal{N}_2(\cdot)$, which gives the complete response surface. The entire framework is depicted in Fig. 2. Once the complete response surface is obtained, multi-objective optimization is performed on the predicted response surface, whose details are discussed in the following subsection.

2.3. Non-dominated sorting genetic algorithm- II

Genetic algorithm (GA) is one of the popular evolutionary algorithms that can provide Pareto-optimal solutions from a population using a single run of GA. There are numerous variants of GA implemented in the literature for multi-objective optimization (Knowles and Corne, 1999; Deb et al., 2000). Among those, the non-dominated sorting genetic algorithm (NSGA) (Deb et al., 2000), is commonly used for multi-objective optimization. In this study, version 2 of NSGA is used, which is often called NSGA-II. The algorithm starts with generating a population of N individuals, as shown in Fig. 3(a). It is noted that each randomly generated individual represents the set of decision variables ($\mathbf{X}$) and is encoded in the form of a chromosome. The values of objective functions are estimated to correspond to each individual. Once function values are obtained, they are classified into a number of non-domination levels, as shown in Fig. 3(b). For example, in Fig. 3(b), X_1 and X_2 are said to be non-dominated because function values (f_1 and f_2) at X_1 and X_2 are conflicting, i.e., one increases, the other decreases. Using this concept, the entire population is categorized into different non-domination levels and ranked accordingly. In Fig. 3(b), it is seen that the solutions corresponding to rank 2 are dominated by rank 1. Similarly, solutions corresponding to rank 3 are dominated by rank 2. Thus, rank 1 solutions become non-dominated by any other solutions, called Pareto optimal solutions, which are stored in every iteration using the elitism operator. Non-dominated solution is not only one criteria to obtain the final Pareto front; the diversity of non-dominated solutions in the population is also checked, which is estimated using crowding distance. The crowding distance is defined as the Euclidean distance between each individual on a front, as shown in Fig. 3(c). Once every individual in a population is ranked and a crowding distance is assigned to each individual, NSGA-II selects the individuals with the lowest rank. If two individuals have the same rank, then the individual with the larger crowding distance is selected as a parent, as shown in Fig. 3(d). Using the parents obtained from the

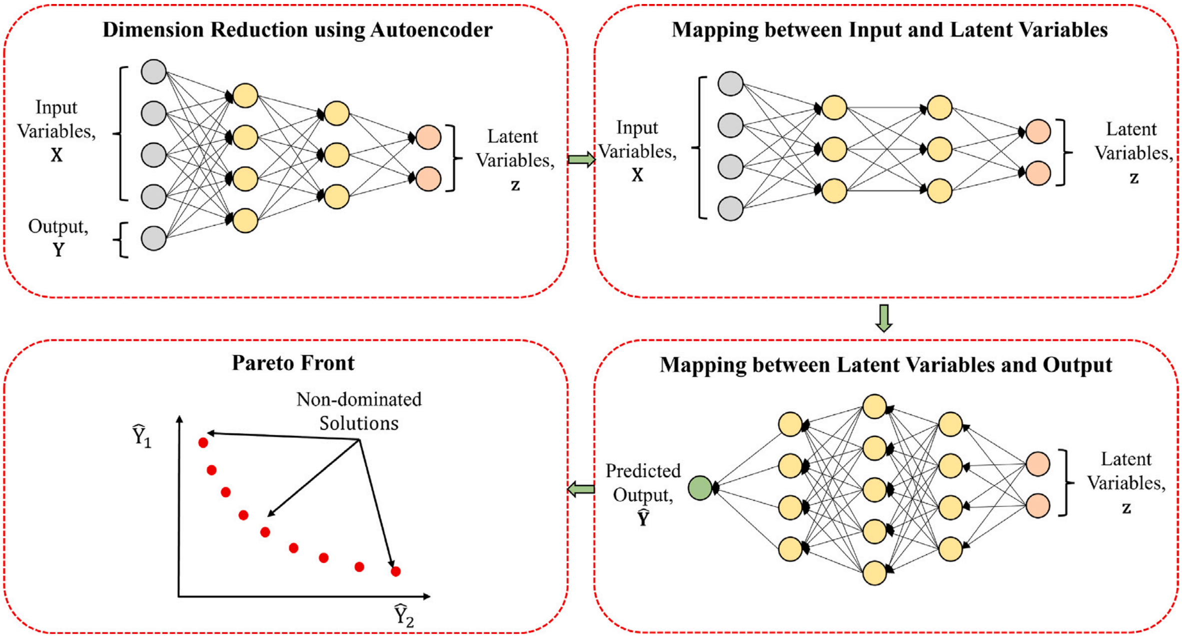


Fig. 2. Schematic diagram for the proposed deep learning-based multi-objective optimization framework.

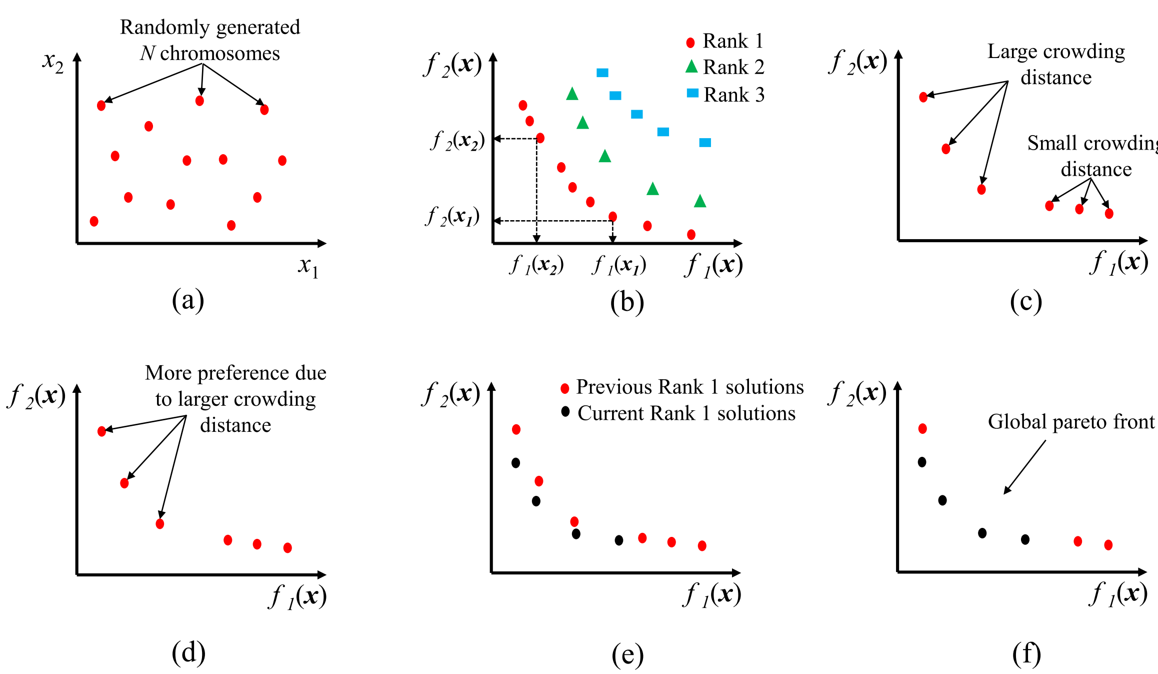


Fig. 3. Schematic diagram for NSGA-II.

current iteration, offspring are generated using crossover and mutation operators (Golberg, 1989). The generated offspring and parent population are combined and used as a population for the next iteration, as shown in Fig. 3(e). The best solutions are selected based on rank and crowding distances. The entire procedure is continued until a specified number of generations is achieved. The final Pareto front is shown in Fig. 3(f).

3. Building model and seismic hazard details

The multi-objective optimization discussed in the previous section is adopted in a CLT-CW system. As described in Section 1, this study builds upon the foundational works previously conducted by the authors (Tesfamariam et al., 2021b; Teweldebrhan and Tesfamariam, 2022; Teweldebrhan et al., 2022, 2023). This section provides a concise summary of the building's structural outline, the methodology employed in the seismic design and numerical model, and the rationale for GM selection.

3.1. Building information

The building considered is a 20-story residential structure, featuring three CLT shear-walls that are coupled with steel coupling beams in every floor and are attached to the base using BRB hold-downs, as illustrated in Fig. 4(a). The first story of the structure has a height of 3.2 m, with the subsequent stories each having a height of 2.96 m. The building's plan dimension is 33 m by 33 m. In the CLT-CW

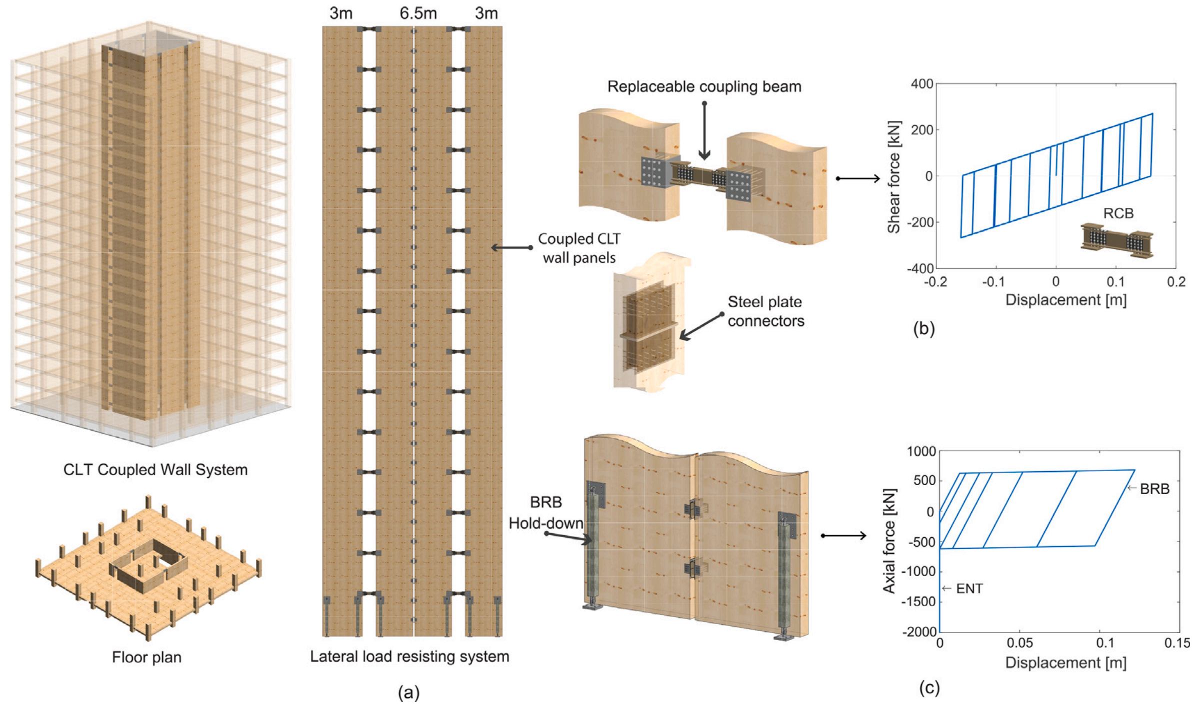


Fig. 4. Building model: (a) CLT-CW system, (b) hysteric response of coupling beam, and (c) hysteric response of BRB hold-down (Tesfamariam et al., 2021a).

configuration shown in Fig. 4, both the end shear walls are 3 m in length, while the central CLT shear wall spans 6.5 m, with all walls maintaining a uniform thickness of 245 mm. Within the CLT-CW system, the components responsible for energy dissipation are the steel coupling beams and the BRB hold-downs. The steel coupling beams (1 m in length) entail a central "fuse" shear link at both ends, connected to steel beam segments. As a secondary energy dissipation component of the CLT-CW system, the BRB hold-downs demonstrate high seismic performance with substantial ductility and energy dissipation. When subjected to lateral force, the wall engages in a rocking motion, effectively transferring axial load to the hold-downs.

3.2. Design methods

In designing the CLT-CW system, four load cases were taken into account: dead load, live load, snow load, and earthquake load. The maximum load, derived from the combination of these cases, was used to design the system's elements. The hypothetical location for the system was Vancouver, BC, and the corresponding snow and seismic loads were established following the standards set by the NBC (2020). The CLT floor panels were designed as one-way slabs using 5-ply CLT of Grade E1, as specified by CSA 086-19 (2019). Subsequently, the gravity load system was analyzed in ETABS, and the elements were designed in accordance with the CSA 086-19 (2019) standard.

For seismic design, both linear and nonlinear analysis models can be applied to CLT-CW systems (Teweldebrhan and Tesfamariam, 2022). One of the pivotal steps in the seismic design process of the CLT-CW systems is selecting an appropriate coupling ratio (CR) value, which significantly affects the structural behavior of the systems. This CR value quantifies the converted base moments in individual walls to axial forces via shear forces, a transformation resulting from the coupling action (Harries and McNeice, 2006). Mathematically, CR is defined as (El-Tawil et al., 2009)

$$CR = \frac{NL_w}{M_{total}} = \frac{NL_w}{NL_w + \sum M_{wall}} = \frac{L_w \sum V_{beam,i}}{NL_w + \sum M_{wall}}, \quad (4)$$

where M_{total} = total overturning moment due to lateral forces, N = axial load induced by the shear forces of the coupling beams, L_w = distance between the centroid of two walls, M_{wall} = reduced base moment of the individual walls, and $\sum V_{beam,i}$ = accumulation of coupling beam shears acting at the edge of all piers.

Based on the CR value, studies categorized the behavior of CW systems into low, intermediate, and high, correlating to the coupling effect of the coupling beams (El-Tawil et al., 2009). Systems with low CR values typically provide limited structural benefits, as they minimally reduce wall moments and lateral drifts. Conversely, higher CR values result in increased axial forces within the wall piers. This augmentation in axial forces can lead to premature crushing failures at the base of the walls (El-Tawil et al., 2002). An intermediate CR value (30% to 45%) is the most optimal and recommended choice (El-Tawil et al., 2009). This range effectively balances the structural benefits without inducing detrimental effects. In line with this understanding, the authors have explored the impact of varying CR values from 10% to 50% on the structural response of CLT-CW systems. In their exploration, detailed in previous research (Teweldebrhan et al., 2023), the authors identified a CR value of 30% as the optimal choice for balancing structural integrity and resiliency. Accordingly, in this study, three baseline systems with CR values of 30% were adopted. The targeted CR value can be achieved employing various coupling beam shear profiles (Teweldebrhan et al., 2023). The continuous medium method (CMM) (Smith and Coull, 1991) was utilized to derive nonlinear shear strength demands in the coupling beams, considering uniform (an average of total shear strength demand) or stepped coupling beam strengths. These three profiles of coupling beam shear formed the deterministic baseline cases. Subsequently, seismic demands in each component, such as CLT balloon shear walls, steel coupling beams, and hold-downs, were determined and incorporated into their design.

3.3. Numerical model

A two-dimensional (2D) numerical model of the systems is developed in Open System for Earthquake Engineering Simulation (OpenSees) (McKenna et al., 2000). The model is constructed by assembling the CLT shear-wall elements, coupling beams, and hold-downs. The CLT wall panels, characterized by their high elastic stiffness, behave as a rigid body during their in-plane responses. Thus, the panels were

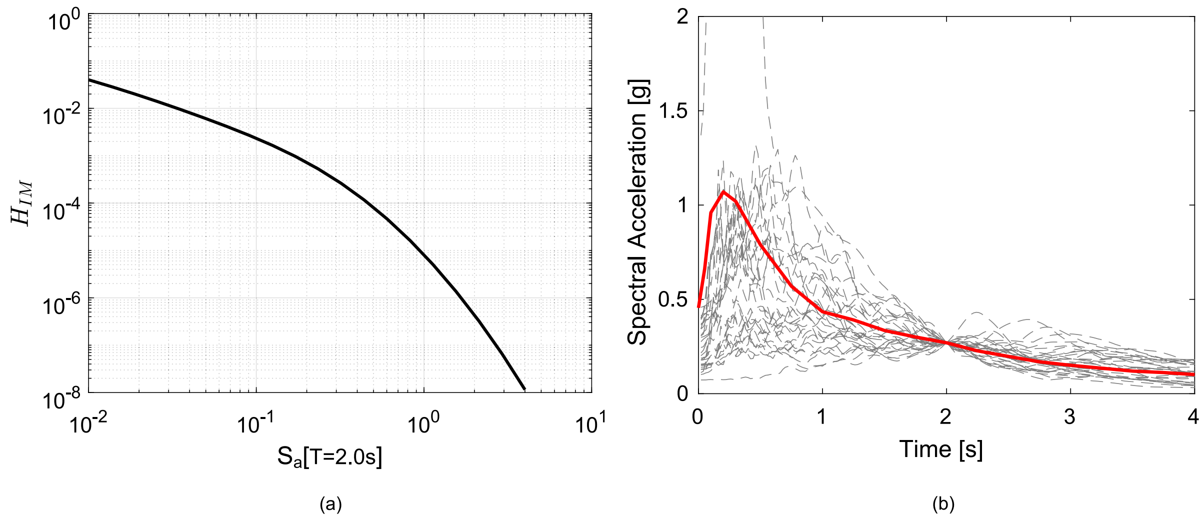


Fig. 5. GM selection: (a) seismic hazard curve, (b) response spectra for scaled GM records.

modeled as a single equivalent layer using OpenSees *ElasticIsotropic* material and *quad* element (Demirci et al., 2018). A thickness of 245 mm is employed with a Poisson's ratio of 0.3 and an equivalent Young's modulus of 10.54 GPa. To simulate the continuity of the CLT wall panels characteristic of balloon systems, *ZeroLength* vertical and horizontal spring elements are utilized using *uniaxialMaterial* elastic materials (Teweldebrhan et al., 2022). Steel coupling beams with central shear links demonstrate stable and ductile behavior under cyclic shear loading. While the central fuse element exhibits a bilinear response under lateral loads, the end beams remain elastic. To simulate the response of the coupling beam (Fig. 4(b)), the OpenSees *Steel01 UniaxialMaterial ZeroLength* non-linear spring and *elasticBeamColumn* elements were utilized to represent the shear-link and end beams, respectively. Parameters for the shear-link model, including the initial elastic and post-yielding stiffness, were adapted from Zona et al. (2018) and Ji and Molina Hutt (2020). Acknowledging the findings from prior studies that identify rocking as the dominant deformation mode in balloon CLT shear wall systems (Li et al., 2022; Tesfamariam et al., 2021a), the BRB hold-downs were modeled using OpenSees *Steel01 UniaxialMaterial* with *ZeroLength* axial springs at the wall corners, and to represent the base compression interaction, *uniaxial elastic no-tension (ENT)* material was utilized parallel to the BRB, as depicted in Fig. 4(c) (Tesfamariam et al., 2021a). The model parameters for this element were sourced from Tesfamariam et al. (2021a) and were based on data from CoreBrace, the manufacturer of the BRB braces. Moreover, horizontal stiff springs were also incorporated to restrain the lateral sliding at the base. The vertical loads equivalent to the gravity loads carried by the gravity columns were depicted through P-Delta effects accounted for in a leaning column modeled using *elasticBeamColumn* elements. Elastic *UniaxialMaterial* and *truss* elements were employed to link the CLT-CW system and leaning column and transfer the P-delta effect. The cross-sectional areas of the leaning column and trusses were assigned to be large enough to remain axially rigid.

3.4. Eigen value analysis and ground motion selection

After assembling and modeling the structural components in OpenSees, eigenvalue analyses were conducted. These analyses enabled the determination of the first three fundamental periods of the CLT-CW systems. The first fundamental periods of the CMM, uniform, stepped, and optimal systems were recorded as 1.88, 1.87, 1.89, and 1.89 s, respectively. Accordingly, 50 ground motion GM records (bi-directional) were selected using an in-house probabilistic seismic hazard analysis (PSHA) tool incorporating the NBC 2020 6th generation seismic hazard model (Tesfamariam et al., 2023). The primary inputs for this tool were the return period (T_R), desired GM record count, anchor period (T_A), and the range of periods to match response spectral values (T_{min} to T_{max}). The latter was defined by the higher mode periods of the structural systems. Using $T_R = 2475$ years and $T_A = 2.0$ s, the PSHA was conducted to provide the necessary site-specific hazard information in terms of magnitude, distance, and seismic hazard contributions. Three seismic sources – crustal, in-slab, and interface – significantly contribute to seismic hazards in Vancouver, BC (Goda and Atkinson, 2011). As a result, three target response spectra were established for each earthquake type using the conditional mean spectrum. From the PSHA seismic disaggregation findings, the necessary number of GM records for each source was identified. Records for crustal and in-slab were procured from the PEER-NGA-West of the USA database, while interface records came from the KiK-net of Japan database (Tesfamariam et al., 2023). The same selection method was successfully implemented to evaluate the seismic performance of timber wall-frame systems in Tesfamariam and Teweldebrhan (2023), Teweldebrhan and Tesfamariam (2023b). Figs. 5(a) and 5(b) illustrate the seismic hazard and the comparison of the response spectra of the selected GM records with the target spectrum at $T_A = 2.0$ s, respectively.

4. Problem formulation

Once the combined structural system and GM have been specified, as discussed in Section 3, it is necessary to design the parameters related to coupling beams and hold-down before installation to maintain satisfactory performance while it is exposed to an uncertain environment. It is seen in Fig. 4(a) that the structure is symmetrical. Therefore, twenty coupling beams and three hold-downs are considered to be the design elements of the 20-story building. The numerical model is developed in OpenSees, as discussed in Section 3.3. The yield force and yield displacement are taken as the design variables for each coupling beam and BRB hold-downs. Therefore, there are 46 design variables (i.e., 40 variables for 20 coupling beams and 6 variables for three hold-downs). The optimization problem is formulated with the following objective functions:

$$\min \begin{cases} Y_1 = f_1(\mathbf{X}) = \max\limits_{k\in[1,N]}\left(\frac{1}{N_{sim}}\sum_{i=1}^{N_{sim}} \mathrm{ISDR}_{k,i}\right) \\ Y_2 = f_2(\mathbf{X}) = \max\limits_{k\in[1,N]}\left(\frac{1}{N_{sim}}\sum_{i=1}^{N_{sim}} F^{\max}_{\mathrm{CB},k,i}\right) \\ Y_3 = f_3(\mathbf{X}) = \max\limits_{l\in[1,N_H]}\left(\frac{1}{N_{sim}}\sum_{i=1}^{N_{sim}} F^{\max}_{\mathrm{HD},l,i}\right) \end{cases} \tag{5}$$
$$\text{s.t} \quad \mathbf{X}^{(L)} \leq \mathbf{X} \leq \mathbf{X}^{(U)}$$

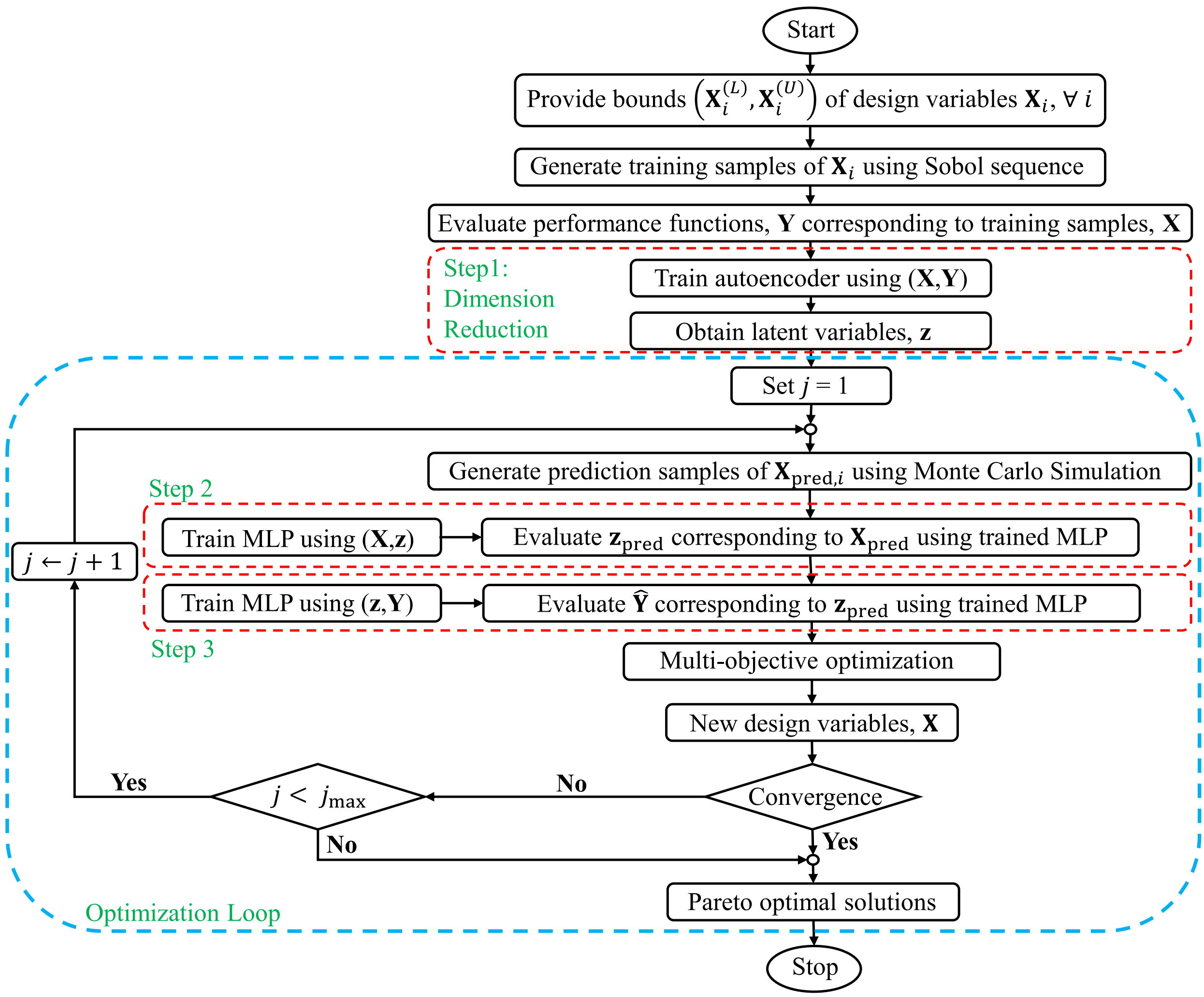


Fig. 6. Flowchart for proposed deep learning based multi-objective optimization.

where ISDR, $F_{\text{CB}}^{\max}$, and $F_{\text{HD}}^{\max}$ denote the interstory drift ratio, the ultimate force in the coupling beam at a given story, and the ultimate force in the BRB hold-down, respectively. Also, the design variables are represented by $\mathbf{X} = \left[F_{\text{CB},1}^{y} \quad u_{\text{CB},1}^{y} \quad \cdots \quad F_{\text{CB},N}^{y} \quad u_{\text{CB},N}^{y} \quad F_{\text{HD},1}^{y} \quad u_{\text{HD},1}^{y} \quad \cdots \quad F_{\text{HD},N_H}^{y} \quad u_{\text{HD},N_H}^{y}\right]^T$. The yield forces at a coupling beam and a BRB hold-down are denoted by $F_{\text{CB},k}^{y}$ and $F_{\text{HD},l}^{y}$, respectively. Similarly, the yield displacement of the coupling beam and hold-down are represented by $u_{\text{CB},k}^{y}$ and $u_{\text{HD},l}^{y}$, respectively. In Eq. (5), N, and N_H represent the number of stories and the number of BRB hold-downs, respectively. The total number of ground motions is represented by N_{sim}. The multi-objective optimization framework is provided in Fig. 6.

5. Results and discussion

The coupling beams and hold-downs are designed by minimizing the three objective functions, i.e., the maximum inter-story drift ratio (MaxISDR), the maximum ultimate force in the coupling beam among all floors, and the maximum ultimate force in BRB hold-downs. The three objective functions considered in this study are conflicting in nature with each other; thus, a Pareto front is obtained. The optimization is performed by considering an ensemble of GMs, whose details are provided in Section 3.4. The mean values of design variables are taken as follows: $\left[F_{\text{CB},1}^{y} \quad F_{\text{CB},2}^{y} \quad \ldots \quad F_{\text{CB},20}^{y}\right]$ = [21, 40, 57, 71, 83, 94, 102, 109, 114, 118, 120, 122, 122, 122, 122, 121, 119, 118, 117, 117] kN, which is calculated based on CMM (Smith and Coull, 1991; Teweldebrhan and Tesfamariam, 2022); the mean value of yield displacement of the coupling beams in all floors, $u_{\text{CB},i}^{y}$ is assumed to be 1 mm (i= 1, …,20) (Zona et al., 2018; Teweldebrhan and Tesfamariam, 2022); the mean value of yield force of the three BRB hold-downs, $\left[F_{\text{HD},1}^{y} \quad F_{\text{HD},2}^{y} \quad F_{\text{HD},3}^{y}\right]$ = [1643, 1643, 4948] kN, and the mean value of yield displacement of all hold-downs is kept the same, whose value is 4.4 mm (Teweldebrhan et al., 2023). The design variables in this study are assumed to adhere to a uniform distribution with a coefficient of variation of 20%. As a result, the problem under consideration consists of 46 design variables. Importantly, the average yielding forces for the coupling beams and hold-downs were obtained from a CLT-CW system designed with a CR value of 30%.

The proposed deep learning-based multi-objective optimization starts with the generation of training samples of the design variables. To do that, 150 samples of each design variable are generated using the Sobol sequence, a Design of Experiment scheme, based on the lower and upper bounds of the design variables. The bounds are estimated as $\mathbf{X} \in \left[\mu_{\mathbf{X}} - \sqrt{3}\sigma_{\mathbf{X}} \quad \mu_{\mathbf{X}} + \sqrt{3}\sigma_{\mathbf{X}}\right]$ where $\mu_{\mathbf{X}}$ and $\sigma_{\mathbf{X}}$ are the mean and standard deviation of the design variable, respectively. The objective functions as defined in Eq. (5) are estimated to correspond to each training sample by performing time-history analysis.

As mentioned in Section 2.2, firstly, the autoencoder is used to reduce the dimension of the problem. It is noted that the accuracy of the proposed method is highly dependent on the accuracy of the autoencoder. Therefore, a comparative analysis is performed to find the dimension of the problem where the mean square error (MSE) between

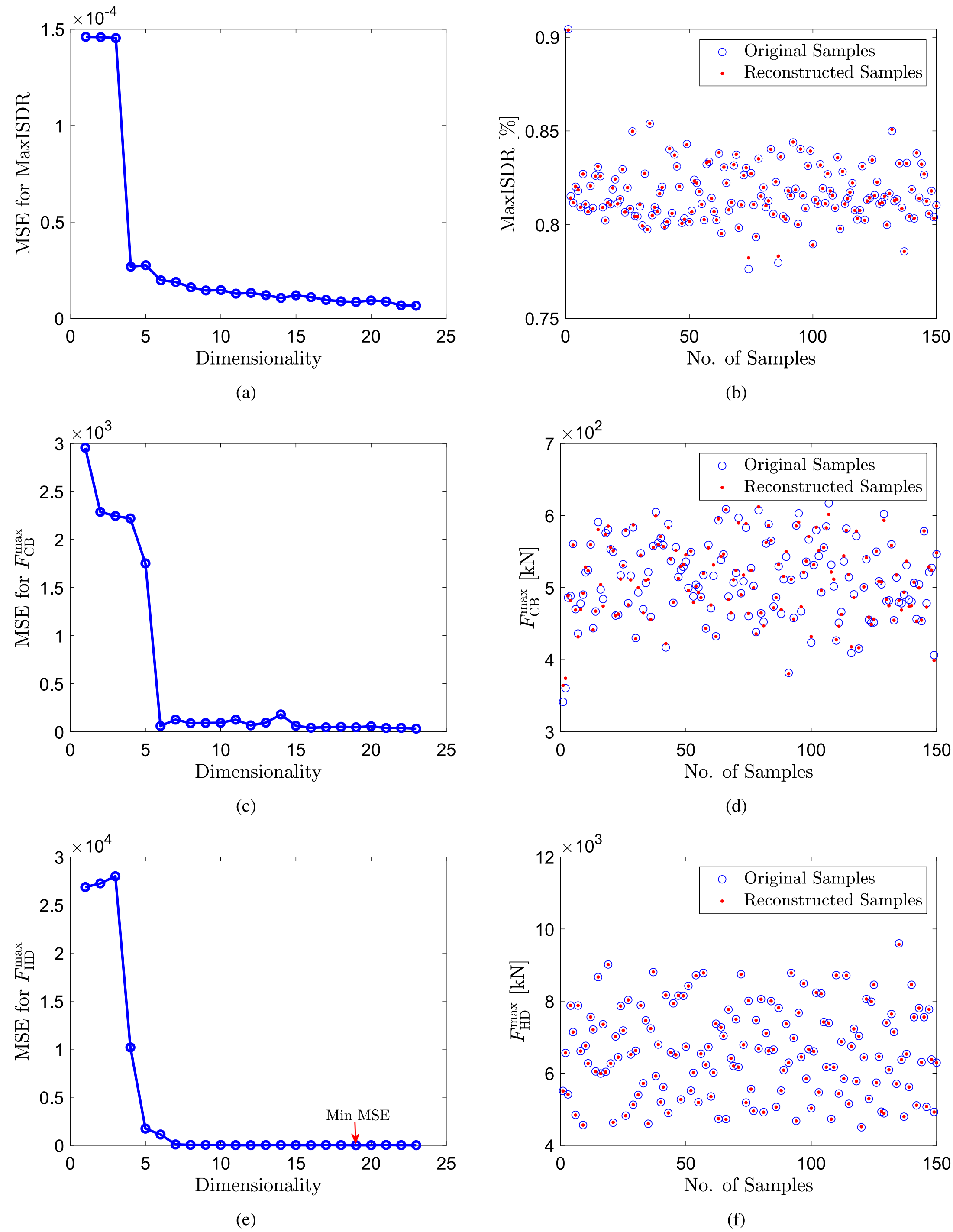


Fig. 7. MSE values corresponding to different size of latent variables for (a) Interstory drift ratio, (c) Ultimate force in coupled beam and (e) Ultimate force in hold-down; The original and reconstructed samples at the dimension where MSE is minimum for (b) Interstory drift ratio, (d) Ultimate force in coupled beam and (f) Ultimate force in hold-down.

the original samples and the reconstructed samples obtained from the autoencoder is the minimum. The autoencoder is trained by using the training design variables and corresponding responses obtained from time-history analysis, and the size of the latent variables is increasing from 1 to 23 with an increment of 1. Here, the log-sigmoid function is used as the activation function for each neuron. Fig. 7 shows the MSE values between original and reconstructed samples obtained from the autoencoder while the size of latent variables is increased. Due to the paucity of space, only the output responses (i.e., three objective functions) are shown here. Fig. 7(a) shows the MSE values for the maximum interstory drift ratio, in which it is seen that when the size of latent variables is 23, the MSE value is minimum. The original and reconstructed samples for interstory drift ratio are shown in Fig. 7(b). Similarly, the size of latent variables for the other two objective functions, i.e., ultimate force in the coupling beam and hold-down, are obtained as 23 and 19, respectively, where MSE values are minimum. Figs. 7(c) and 7(e) show the MSE values for the ultimate force in the coupling beam and hold-down, respectively, and the corresponding reconstructed samples are shown in Figs. 7(d) and 7(f), respectively.

Once the size of the latent space is obtained, we are focused on establishing a mapping between the input variables and the latent variables. Therefore, a deep neural network is used, which is trained

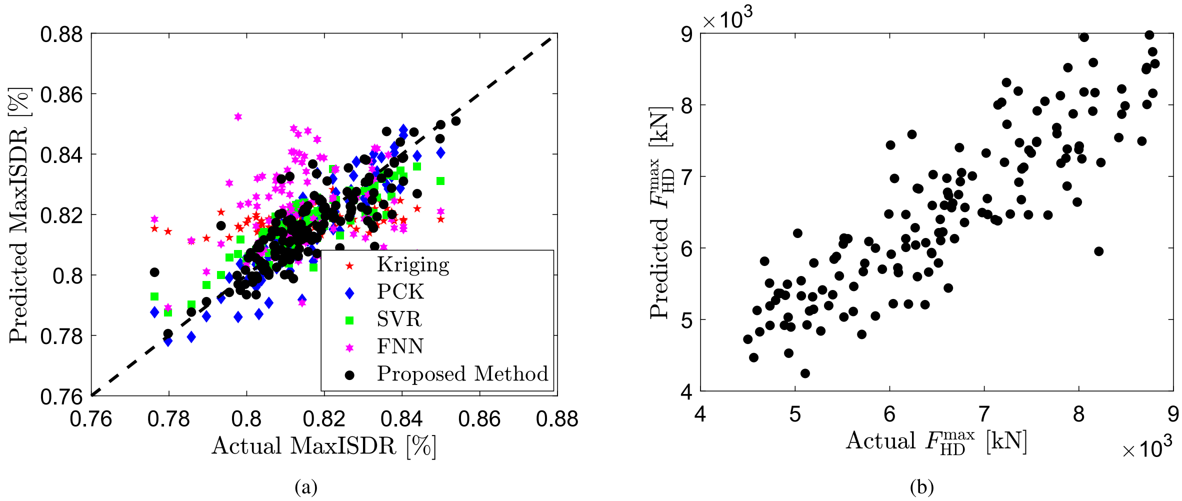


Fig. 8. Scatter plots showing the performance of the proposed method for the predicted responses of (a) Maximum interstory drift ratio and (b) Maximum ultimate force in BRB hold-downs.

by the samples of input variables and latent variables obtained from the encoder part of the autoencoder. Once the training is completed, a large number of samples of the input variables ($\mathbf{X}_p$) are generated using Monte Carlo simulation and passed through the trained neural network, which gives the predicted values of the latent variables, $\mathbf{z}_p$ corresponding to $\mathbf{X}_p$. The architecture of the network is set to two hidden layers, with 20 neurons in each hidden layer. Also, the tangent hyperbolic function is used as an activation function in each neuron. To estimate the weight and bias of this network by minimizing the mean square error loss function, the ADAM optimization algorithm is used in this study. Once a mapping between the input variables and latent variables is established, another neural network is considered to form a mapping between the latent variables and output responses. This network is trained using the latent variables obtained from the encoder and the output responses corresponding to the input variables used to train the previous neural network. Here, the architecture of the network is set to two hidden layers with 10 neurons in each hidden layer. Once the training of this network is completed, the predicted values of latent variables ($\mathbf{z}_p$) obtained from the previous network are passed through this network, resulting in the predicted output responses. As it is seen, the accuracy of the proposed framework is highly dependent on the accuracy of the neural networks. If the neural network is not accurate enough, the results obtained will be erroneous. Therefore, it is important to first illustrate the accuracy of the neural networks before doing the optimization. Fig. 8 shows the scatter plots showing the actual output and predicted output obtained from the proposed framework. The predicted values of the maximum interstory drift ratio is shown in Fig. 8(a). The coefficient of determination (R^2) for the maximum interstory drift ratio is found to be 0.93. To establish the efficiency of the proposed framework, its performance is compared with surrogate models such as Kriging, polynomial chaos-Kriging (PCK), support vector regression (SVR), and feed-forward neural network (FNN), as shown in Fig. 8(a). The corresponding R^2 values are obtained as 0.15 for Kriging, 0.78 for PCK, 0.74 for SVR, and 0.81 for FNN. It is seen that the proposed framework used in this study has a higher R^2 value compared to the traditional machine learning method. Similarly, the R^2 value for the maximum ultimate force in BRB hold-downs is found to be 0.89 using the proposed framework, which is shown in Fig. 8(b).

Having established the accuracy of the proposed framework, multi-objective optimization is performed based on the output responses obtained from the proposed framework. The objective functions considered in this study are described in Section 4. Here, the non-dominated sorting genetic algorithm (NSGA)-II is used for optimization purposes. The parameters related to NSGA-II, i.e., crossover percentage, mutation percentage, and mutation rate, are chosen as 0.7, 0.4, and 0.02,

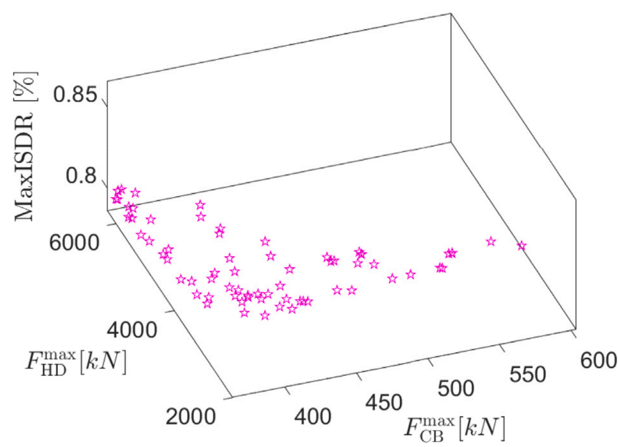
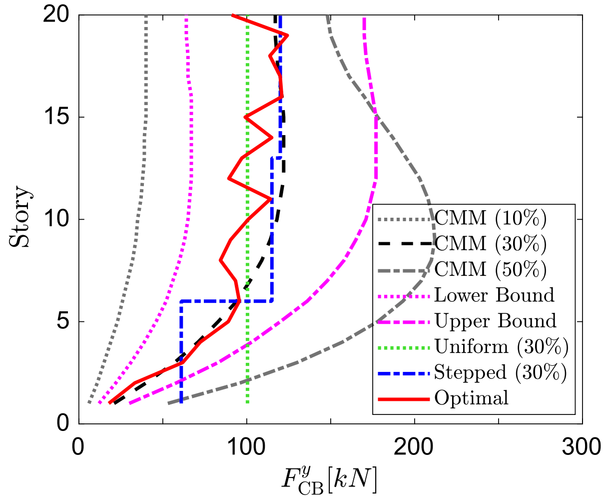


Fig. 9. Pareto front obtained using proposed framework.

respectively (Das and Tesfamariam, 2022). Fig. 9 shows the non-dominated solutions obtained from the proposed deep learning-based multi-objective optimization, which forms a Pareto front. The optimal values for the maximum yield force in the coupling beams across all floors are depicted in Fig. 10(a). For context, seven other profiles are also showcased. The coupling beam shear force distributions for a CR of 10% and 50% delineate the outer bounds of the figure. These distributions represent the lowest and highest CR values feasibly applicable to the CLT-CW system (Teweldebrhan et al., 2023), respectively. In the same study, a system with a CR of 30% was determined to be optimal and thus served as the baseline for this study. This baseline was constructed using three distinct shear force distributions – CMM (30%), uniform (30%), and stepped (30%) – all of which are illustrated in Fig. 10(a). Additionally, the figure also displays the lower and upper bounds of the optimization problem, which were derived from the CMM (30%) distribution. Notably, it can be concluded that the optimal system (obtained in this study) has a CR value ≈ 28%, closely aligns with the CMM (30%) profile in the bottom 7 and top 4 story levels, and exhibits relatively lower demands at intermediate story levels. On the other hand, Fig. 10(b) summarizes the distribution of shear displacement responses of the different CLT-CW systems. As can be seen from the figure, the system with CR = 10% exhibited a higher mean response with a larger dispersion. The system with a higher CR (= 50%) showed a relatively lower mean shear response. The optimal and three baseline systems with CR = 30% (CMM, uniform, and stepped)



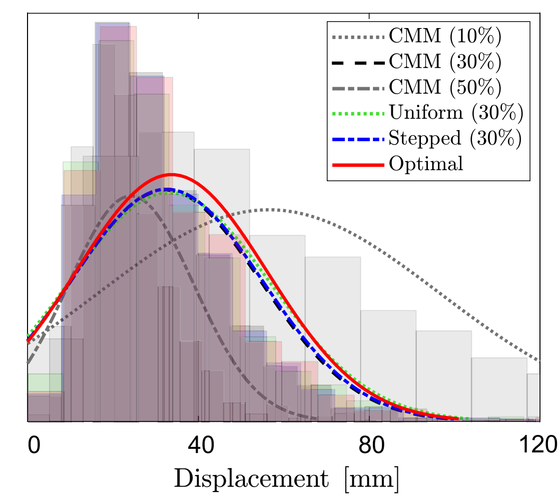


Fig. 10. Coupling beam: (a) shear force profiles and (b) density plots of maximum shear displacement responses.

exhibited almost the same median response, with the optimal design exhibiting some robustness compared to the deterministic baseline systems.

The axial force demand and displacement response of the BRB hold-downs are detailed in Fig. 11. Within Fig. 11(a), the peak yield forces in three specific BRB hold-downs (HD_1, HD_2, and HD_3) are presented across the different distinct systems: CMM (10%), CMM (30%), CMM (50%), and the optimal variant. Also displayed are the lower and upper bounds derived for the optimal system based on the CMM (30%) profile. It is evident that the axial demands in the outer hold-downs (HD_1 and HD_2) are influenced by both flexural action and axial force stemming from the coupling action. An increase in CR values directly amplifies this axial demand. In contrast, the axial demand on the central shear wall solely arises from flexural action due to the neutralization of coupling forces from the two adjacent walls. Therefore, an increase in CR value correspondingly reduces the flexural moment and axial demand on this hold-down (HD_3), refer to Fig. 11(a). The optimal yield forces obtained by the proposed framework have a unique distribution of demands. Even though its lower and upper bounds are defined from the CMM (30%), the determined demand closely aligns with the lowest demands from the three CR values, effectively paralleling the lower bound. Specifically, the yield force demand for HD_1 decreased from 1643 kN to 1523 kN post-optimization, marking a 7.3% reduction. The demands for the HD_2 and HD_3 hold-downs experienced reductions of 17.7% and 31.8%, respectively. Fig. 11(b) provides insights into the density distribution of the axial displacement responses of various CLT-CW systems. As expected, the optimal system has a relatively higher mean/median response, attributed to the minimal strength imparted to the hold-downs. This system also exhibits robustness when contrasted with the deterministic baseline systems. It is important to note that while Fig. 11 represents the demand and response of the hold-down for systems following the CMM method (10%, 30%, and 50%) and the optimal design, the results for the uniform (30%) and stepped (30%) baselines are identical to the system with CMM (30%).

The optimal design obtained in this study not only reduces the demand in the coupling beams and hold-downs, but the system has also exhibited lower top floor displacement when compared with the baseline systems: CMM (30%), uniform (30%), and stepped (30%). Taking a specific GM record into consideration, it is seen that the maximum top floor displacement for the CMM (30%), uniform (30%), and stepped (30%) are found to be 935 mm, 968 mm, and 941 mm, respectively. In contrast, the optimal design yielded a peak displacement of 845 mm, marking an impressive 13% reduction in the top floor's peak displacement. Similarly, the interstory drift ratio for all floors for the baseline and optimal systems is shown in Fig. 12. The optimal system registers a higher interstory drift ratio at the first story, a finding congruent with the higher hold-down axial displacement responses illustrated in Fig. 11(b). The uniform shear force profile system demonstrates an increased interstory drift ratio at the top 5 stories, attributable to the system's reduced strengths at these levels, as detailed in Fig. 10(a) (Teweldebrhan et al., 2023).

The analysis of this study is extended to examine the performance of the optimal CLT-CW systems at the collapse level. The collapse MaxISDR of CLT buildings typically falls within the 4.5 to 5.5% range (Teweldebrhan and Tesfamariam, 2022, 2023a). Prior research on the CLT-CW system considers the collapse threshold at 5% MaxISDR (Teweldebrhan et al., 2022). As a result, an incremental dynamic analysis (IDA) is carried out, progressively scaled GMs intensity. The IDA captures the entire dynamic response of the system, ranging from its elastic behavior to its collapse, as described by Vamvatsikos and Cornell (2002). The chosen intensity measure of the GMs is the spectral acceleration at the system's first fundamental period, denoted as $S_a(T_1)$. Given these parameters, the IDA curves for the optimal system is depicted in Fig. 13(a).

The collapse fragility curve, which illustrates the probability of collapse of the systems at varying seismic intensities, is plotted by fitting the collapse spectral accelerations acquired from the IDA. A log-normal cumulative distribution function (CDF) is utilized to fit the collapse data points, and the method of moments is used to determine the distribution function's parameters (namely, the mean and standard deviation). Fig. 13(b) presents the collapse fragility curves for all the CLT-CW systems. Most of the baseline systems exhibit similar probabilities of collapse curves, though the stepped and CMM systems display marginally better performance. In contrast, the optimal system shows reduced performance at higher ground motion intensities when compared to the baseline systems. This can be attributed to the fact that the system had lower hold-down and coupling beam strengths at the targeted earthquake level while optimizing the system. Low coupling beam strength yields lower CR value. Previous study by Teweldebrhan and Tesfamariam (2022) using IDA have shown that while CLT-CW systems with different CR values perform similarly under low seismic intensities, at higher intensities, systems with lower CR values yield sooner, ceasing coupling action and vice versa. This variation in CR effectiveness at different seismic intensities leads to pronounced performance differences, especially as systems exhibit non-linear responses under more severe seismic conditions. Similarly, the uniformly profiled system also displayed a relatively lower performance (compared to the other two baseline cases) due to it being provided with lower

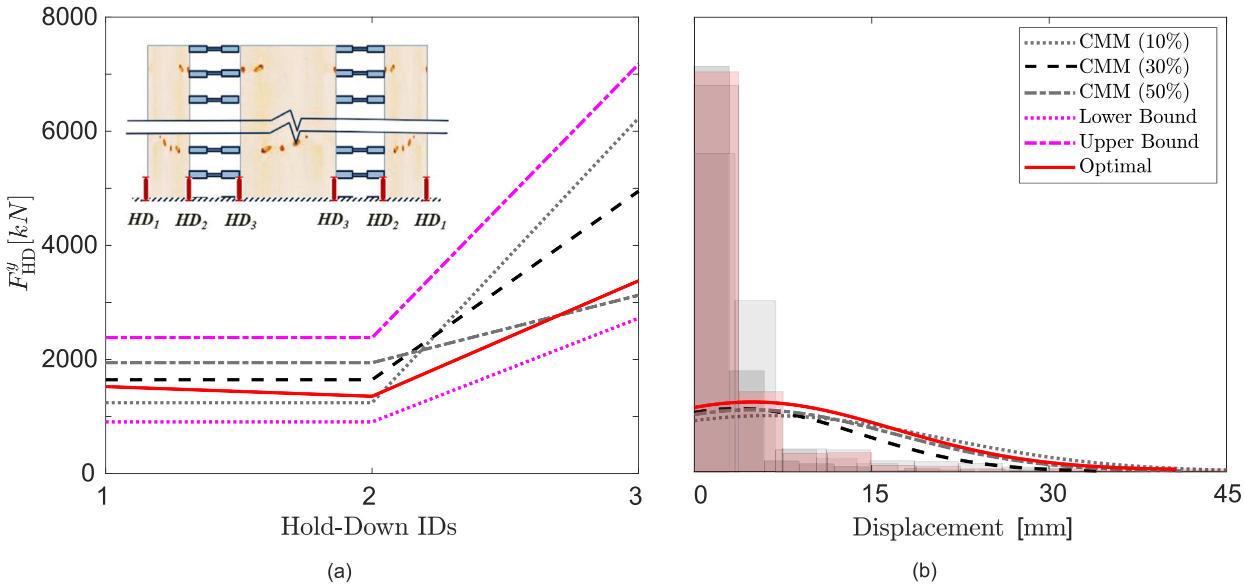


Fig. 11. BRB hold-down: (a) axial force demands and (b) density plots of maximum axial displacement responses.

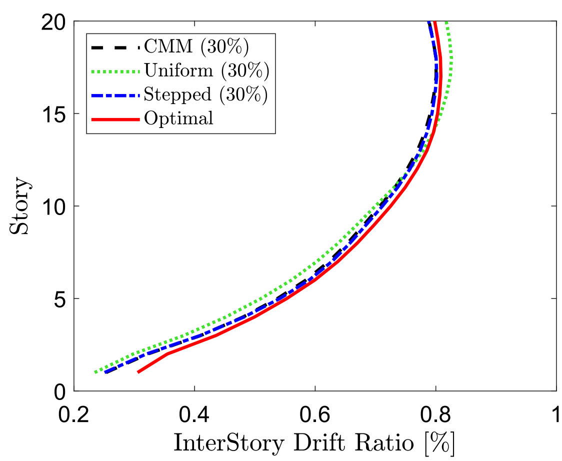


Fig. 12. Mean values of maximum interstory drift ratio of CLT-CW system for 20-story timber building.

coupling beam strengths (than the required) at higher stories, leading to higher responses and residual deformations observed in the coupling beam responses. Detailed probabilistic seismic performance assessment of the systems is performed using multi-variate fragility analysis by the Teweldebrhan et al. (2023). To enhance this analysis and provide a more comprehensive understanding of the uncertainty and variability inherent in the system's response, fragility calculations that include confidence bounds (e.g., Aloisio et al. 2022) could be used to further strengthen the robustness of the seismic performance assessments.

6. Conclusion

In this study, a novel approach has been proposed for high-dimensional multi-objective optimization (MOO) using an autoencoder-based deep learning framework. The traditional evolutionary algorithms for solving MOO problems require multiple computational model evaluations to obtain Pareto optimal solutions, which may lead to an increase in computational cost for complex models. To reduce the cost, surrogate models are useful, which approximate the original cost functions of MOO problems. It is seen that the surrogate models often suffer from the curse of dimensionality when the dimension of the problem is large. To address this, in this study, an autoencoder-based three-step framework is proposed for solving MOO problems. Also, this research marks a significant advancement in timber-based construction, specifically focusing on optimizing the CLT-CW system for tall mass timber buildings. By considering uncertainties in connection elements and applying deep learning-based high-dimensional multi-objective optimization, the study refines the design solutions, enhancing the system's performance under seismic conditions. Three deterministic models developed in OpenSees were examined using non-linear time history analysis to serve as a baseline comparison for the optimal solution. A key achievement for the optimal solution is the adept adjustment of design strengths and the incorporation of uncertainties to get an optimal system, with maximum interstory drift ratio serving as a pivotal performance indicator, supplemented by residual interstory drift ratio and maximum floor acceleration. This work underscores the importance of addressing uncertainties in timber structures and sets a foundation for future studies aiming to develop resilient and sustainable high-rise timber buildings, ultimately contributing to meeting the challenges of urban densification and environmental sustainability. The insights gained from this study can be summarized as follows:

- To establish the efficacy of the proposed framework, its performance is compared with surrogate models such as Kriging, polynomial chaos-Kriging (PCK), support vector regression (SVR), and feed-forward neural network (FNN). The coefficient of determination (R^2) values are 0.15, 0.78, 0.74, and 0.81 for Kriging, PCK, SVR, and FNN, respectively, whereas the same is obtained as 0.93 using the proposed framework.
- For enhanced control and robustness, the connection parameters linked with the coupling beams and hold-downs are fine-tuned using a deep learning-based multi-objective optimization approach. Specifically, the yielding forces and displacements of these two energy dissipative components are optimized.
- The force parameters for the baseline systems were determined using CMM, a method that is both elastic and approximate in nature. Importantly, the post-optimization analysis indicates a marked reduction in force demand when compared to values obtained from CMM. The difference is significant in the hold-down forces. Thus, the optimal system showed a potential enhancement in the reliability of the CLT-CW system.
- The purpose of the proposed high-dimensional multi-objective optimization framework is examined by comparing the structural

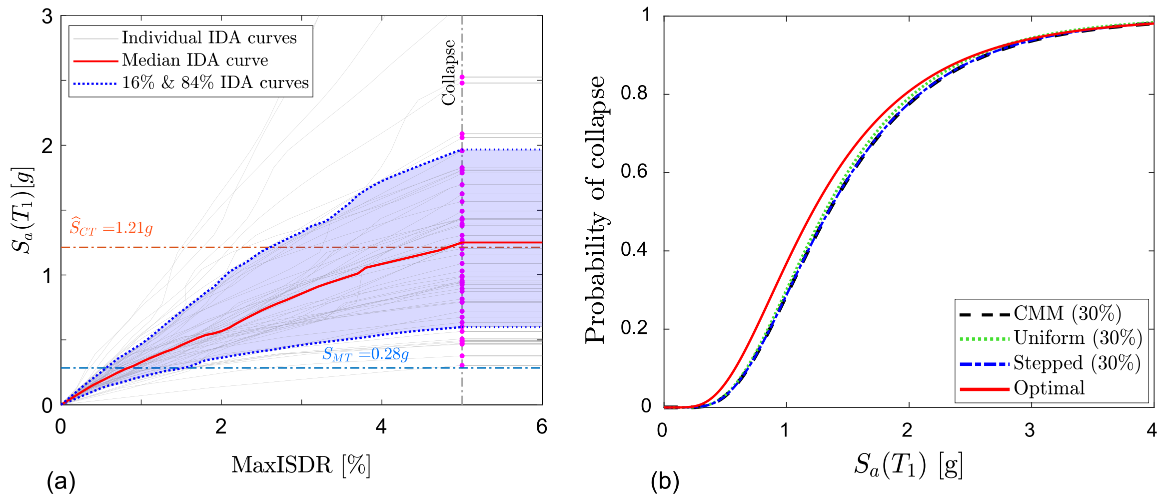


Fig. 13. (a) IDA results for the optimal model and (b) collapse fragility curves.

performance of the optimized CLT-CW system with the three deterministic baseline systems. The optimal system shows superior performance in terms of peak reduction of top floor displacement and residual interstory drift ratio of the building at the maximum considered earthquake (MCE) level.

- Contrarily, at higher GM intensity levels, characterized by nonlinear responses, the baseline systems outperform the optimal design. This divergence can be attributed to the optimized system's reduced strengths in coupling beams and hold-downs, which were specifically tailored for performance at the MCE level.

In summary, the deep learning-based multi-objective optimization framework introduced in this study offers a promising high-dimensional optimal design strategy for tall structures. The outcomes distinctly emphasize the benefits of the revamped coupling beams and BRB hold-downs in mitigating earthquake-induced vibrations in high-rise timber buildings. However, this study has certain limitations. The only sources of non-linearity examined here are the coupling beams and hold-downs. Additionally, large stiffness elastic springs were employed to represent the continuous nature of the balloon panels throughout the building's height. The connections between the CLT-to-hold-down and CLT-to-coupling beam were modeled as rigid connections. To move towards a more engineered solution of the proposed work, the authors also recommend a repetitive analyses and design to establish fragility curves that incorporate confidence bounds. These stated limitations warrant further investigation.

CRediT authorship contribution statement

Sourav Das: Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft. **Biniam Tekle Teweldebrhan:** Formal analysis, Investigation, Methodology, Writing – original draft. **Solomon Tesfamariam:** Conceptualization, Funding acquisition, Resources, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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